

(a) Predict

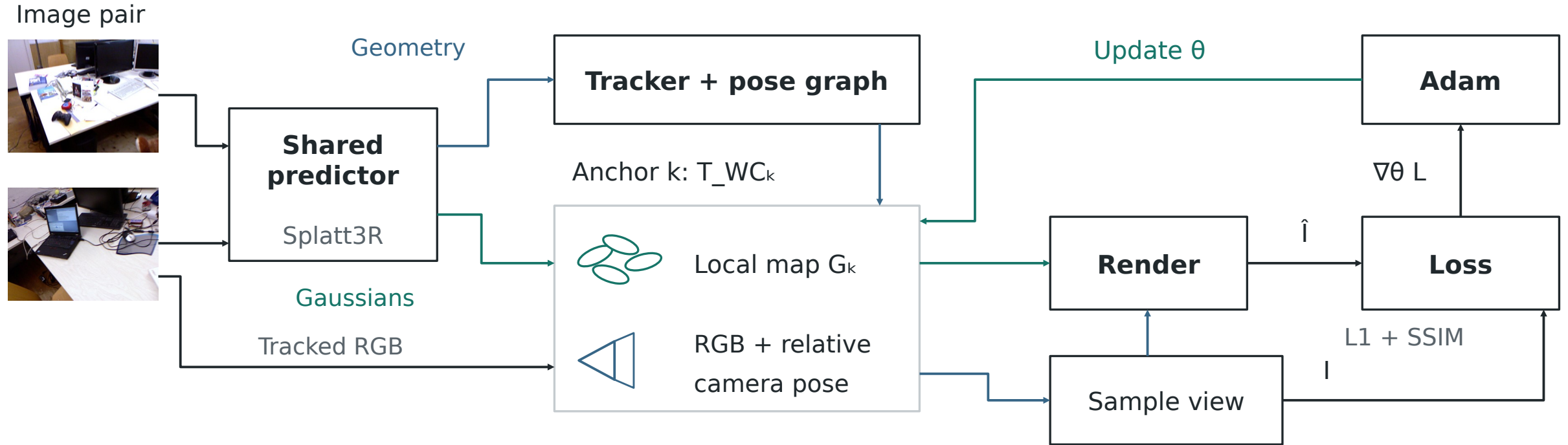
Sec. III-A · Eq. (1)

(b) Anchor

Sec. III-B · Eqs. (2-4)

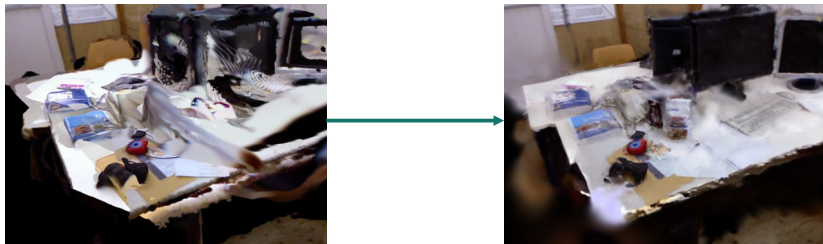
(c) Refine

Sec. III-C · Eq. (5)



Before refinement

After refinement



Shared-anchor relation

$$T_{wcf}^{-1} T_{wck} = T_{ckcf}^{-1}$$

Pose corrections move map and cameras together.

Optional initialisation: Sec. III-D · Eqs. (7-8)

(a) Predict

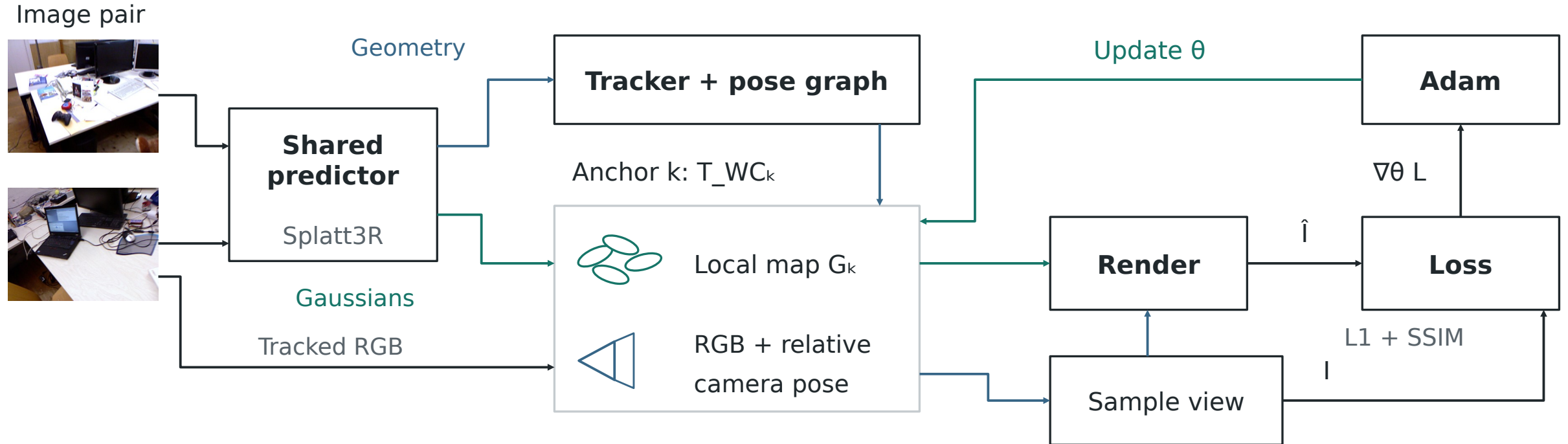
Ch. 2 · Eq. (2.7)

(b) Anchor

Ch. 4 · Eqs. (4.1–4.3)

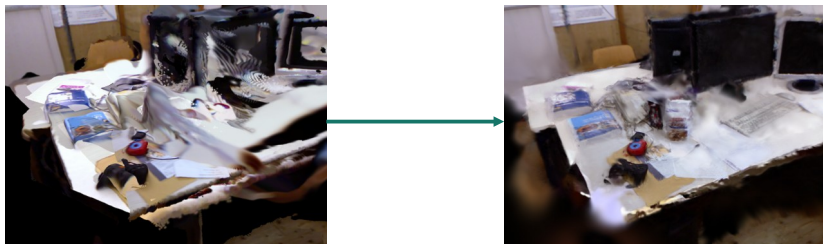
(c) Refine

Ch. 7 · Eq. (7.1)



Before refinement

After refinement



Shared-anchor relation

$$T_{WCf}^{-1} T_{WCk} = T_{CkCf}^{-1}$$

Pose corrections move map and cameras together.

Optional initialisation: Chs. 5–6 · Eq. (6.1)

Why the supervision camera shares the anchor

Coordinate control and refinement loop

Before a loop closure

Local map and camera use keyframe k.
Their relative transform is fixed.



$$T_{WCf}^{-1} T_{Wck} = T_{CkCf}^{-1}$$

Pose correction

The pose graph changes T_{Wck} .
The map and camera move together.



shared anchor: relation preserved

Controlled 6% scale test

Fixed world camera: refinement reverses the correction.
Shared anchor: correction remains after about 500 steps.

Desk + room controls

REFINEMENT LOOP

sample RGB

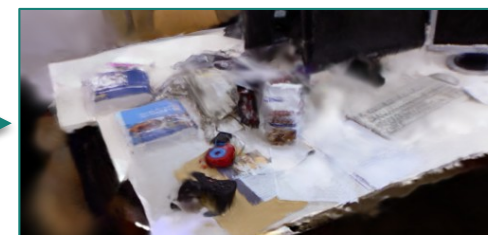
compose pose

render map

L1 + SSIM

Adam update

Only Gaussian scale, rotation, opacity, and DC color receive gradients.



Qualitative reconstruction at common cameras

Measured renders; identical crops and scales

Ground truth

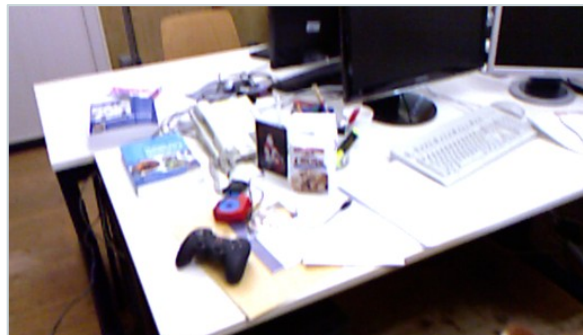
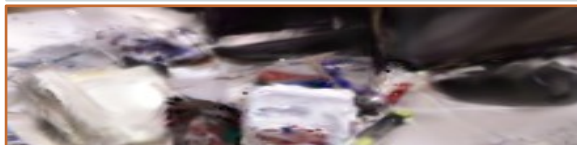
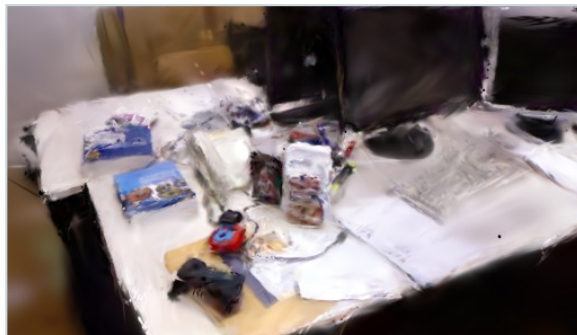
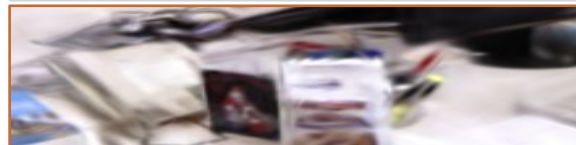
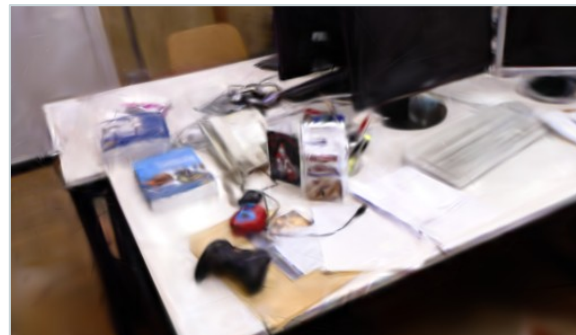


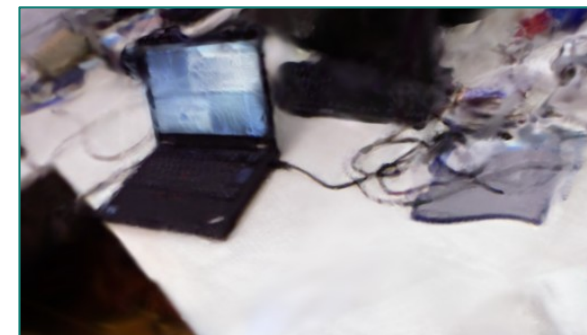
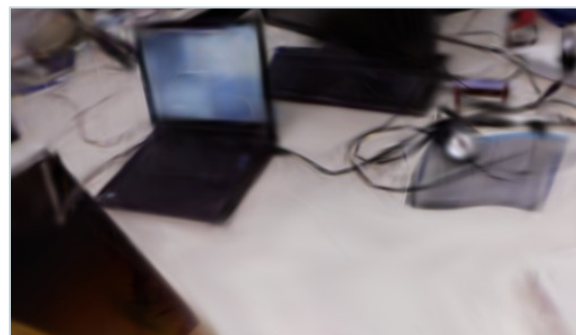
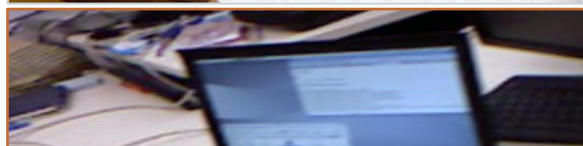
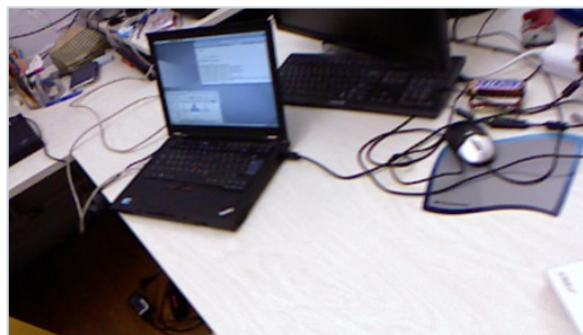
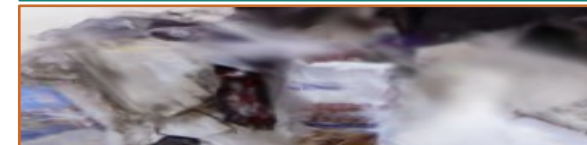
Photo-SLAM



MonoGS



Ours



TUM desk: two middle-ranked Ours-minus-MonoGS PSNR views

Orange panels are identical pixel crops across all methods.

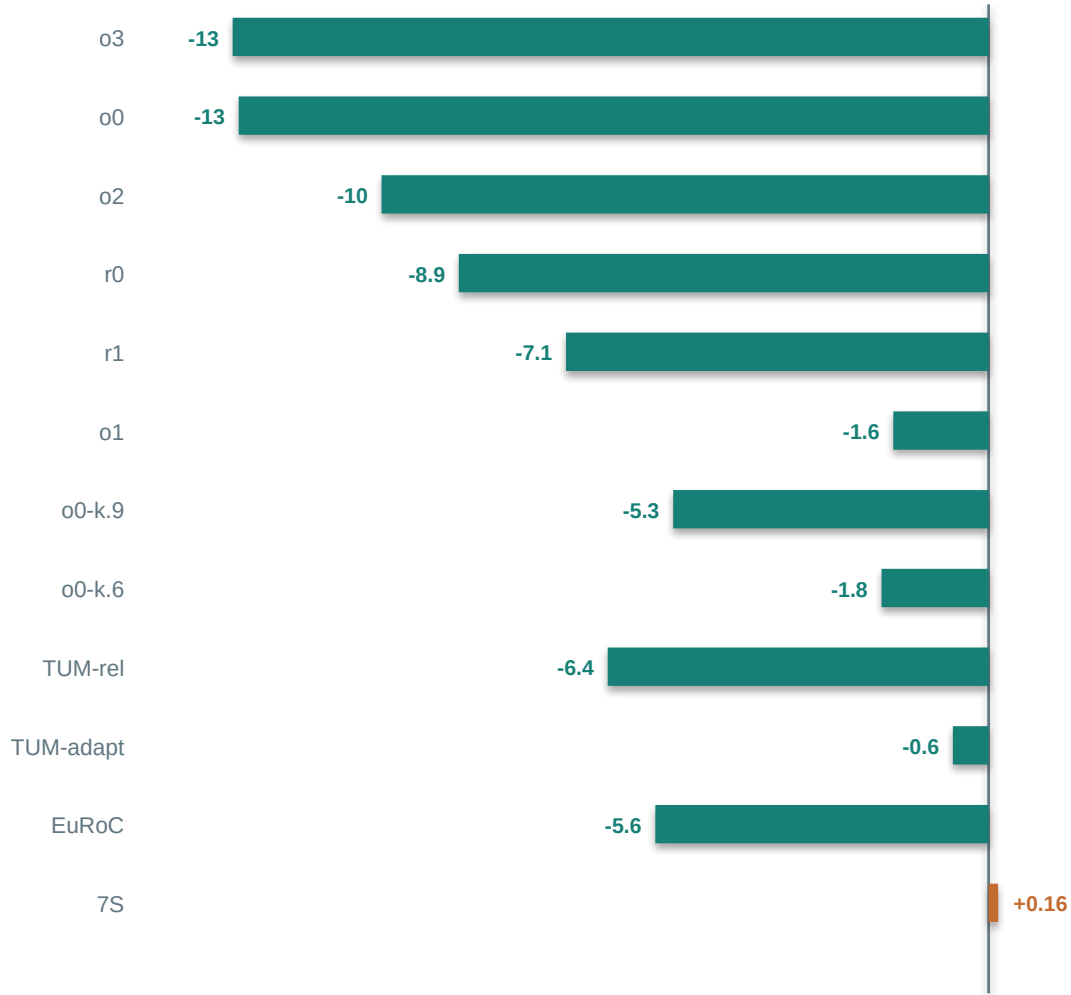
Replica reconstruction across scenes

No monocular MonoGS artifact was available

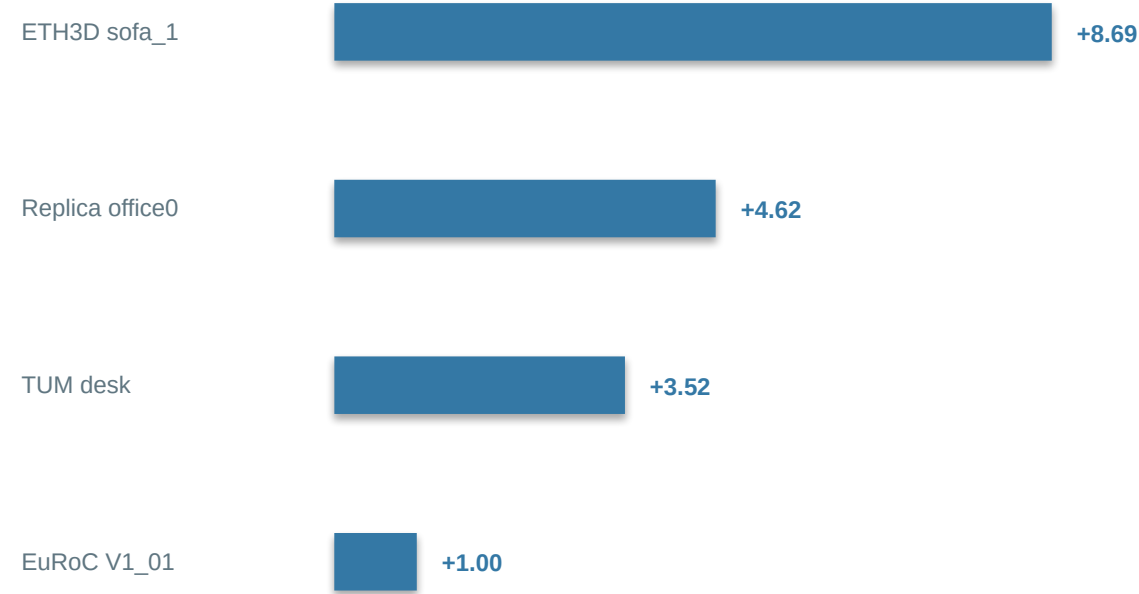


All images use the same selected camera within each scene.

Opacity attenuation: relative LPIPS change (%)



Refiner on vs. off: PSNR gain (dB)



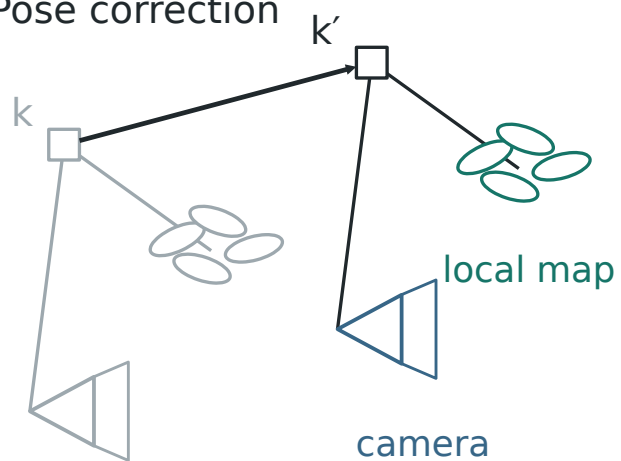
Interpretation

Refinement gives the largest controlled gain.
Opacity reduction is robust; confidence allocation is not.

External baseline comparison uses the released head with attenuation off.

(a) Shared anchor

Pose correction



Map + cameras move together

(b) Same-camera reconstruction

Photo-SLAM



Splatt3R-SLAM



Identical crop · representative office0 view

(c) Replica

Mean PSNR (dB) ↑

